



Wave a Flag

Wave a flag



Easy

General Skills

picoCTF 2021

AUTHOR: SYREAL

Hints ?

Description

1 2 3 4 5

Can you invoke help flags for a tool or binary? This program has extraordinarily helpful information...

warm

194,446 users solved



91%
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 picoCTF{FLAG}

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Flag

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Challenge: Wave a Flag

Category: General Skills

Date: 01/11/26



I. Objective

The objective of this task was to identify the hidden flag embedded within the provided file. The challenge title, "Wave a Flag," together with the description, hints that the solution involves invoking help-related flags on a program or binary. Rather than performing reverse engineering or decoding, the task focuses on executing the program correctly and exploring its built-in help functionality to reveal the flag.

II. Background

In many cybersecurity and capture-the-flag (CTF) challenges, participants are required to interact with unfamiliar binaries using the Linux command line. Unlike Windows, Linux does not rely on file extensions to determine whether a file is executable. Instead, executability is determined by file permissions. Additionally, many Linux programs include built-in help options, commonly accessed using `-h` or `--help`, which display usage information and available arguments. The challenge description explicitly suggests that the program contains "extraordinarily helpful information," encouraging the user to explore these help flags rather than analyze the file's contents directly.

In many digital forensics and cybersecurity challenges, files are intentionally provided without clear identifiers such as file extensions. Operating systems like Windows rely heavily on file extensions to determine how a file should be opened and which application should be associated with it. When an extension is missing, the file may appear unknown or unusable at first glance. However, this does not mean the file's contents are inaccessible. Since all digital files ultimately consist of characters or binary data, they can still be inspected using basic tools. The challenge description stating that the flag was "hidden in plain sight" strongly suggested that the solution involved directly viewing the file's contents rather than transforming or decrypting it.

III. Tool Used

- **Oracle VirtualBox (Kali Linux) App**
- **Linux Terminal**
- **ls** – used to locate the downloaded file
- **wget** – used to download the file directly into the Linux environment
- **chmod +x warm** – used to make the file executable
- **./warm** – used to execute the program
- **./warm -h** – used to invoke the program's help flag



The **-h** or **--help** argument is a common convention in Linux programs to display help messages, usage instructions, or additional information. In this challenge, the help flag was intentionally implemented to reveal the flag itself.

IV. Methodology

1. The file provided in the challenge was downloaded to the local system using the `wget` command, as suggested by the challenge hints.
2. Upon inspection, the file had no extension, making its type unclear in a Windows environment.
3. The file was accessed within a Kali Linux virtual machine, as the program was designed to run only on Linux systems.
4. The program was executed using `./warm` to observe its default behavior.
5. Based on the challenge hint suggesting the use of help flags, the program was executed again with the `-h` argument using `./warm -h`.
6. The help output displayed by the program contained the flag.

V. Result

picoCTF{b1scu1ts_4nd_gr4vy_ac5832c}

```
File Actions Edit View Help
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ ls
flag  warm
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ ./warm
Hello user! Pass me a -h to learn what I can do!
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ ./warm -h
Oh, help? I actually don't do much, but I do have this flag here: picoCTF{b1scu1ts_4nd_gr4vy_ac5832c}
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$
```



VI. Explanation

Upon examining the downloaded file named "warm," Windows was unable to determine its type due to the absence of a file extension. As a result, the operating system could not automatically associate it with a specific program. In Linux, however, file extensions are not required for executables; instead, execution depends on file permissions.

The challenge description emphasized that the program contained **"extraordinarily helpful information,"** and several hints pointed toward invoking help-related arguments. After making the file executable and running it, the use of the **-h** flag proved to be the key step. While not all programs implement help features, this particular binary was designed to reveal the flag through its help output. This approach aligns with the challenge's instructional goal of familiarizing users with Linux command-line conventions and common program arguments.

VII. Conclusion

This challenge highlighted the importance of understanding basic Linux command-line concepts, particularly file permissions and help flags. By following the hints provided and exploring the program's built-in help functionality, the flag was successfully obtained without the need for complex analysis. The task reinforced that careful reading of descriptions and effective use of standard command-line options can often lead directly to the solution.

— **The Analyst: Hyposelenia**